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United States Cost Health Economics & Multi-Payer Revenue Projections for Interventional rTMS Neuromodulation: Multi-State QALYs & Monte Carlo Valuations

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ABSTRACT

Major psychiatric and substance use disorders impose an economic burden exceeding $460 billion annually in the United States, driven by recurrent psychiatric hospitalizations, emergency department over-utilization, and chronic polypharmacy. Repetitive transcranial magnetic stimulation (rTMS) offers an established non-invasive neuromodulation pathway covered under the Centers for Medicare & Medicaid Services (CMS) Physician Fee Schedule. Here we develop a comprehensive health economics and multi-payer revenue projection framework modeling the US healthcare financing architecture across CPT codes 90867, 90868, and 90869, incorporating Geographic Practice Cost Index (GPCI) weights, commercial insurer markups, and Medicaid fee schedules. Coupled with a 5-year Markov cost-utility model, rTMS achieves an incremental gain of 0.606 Quality-Adjusted Life Years (QALYs) per patient and annual direct medical savings of $31,945.00, resultinginanIncrementalCost-EffectivenessRatio(ICER)of$-114,778.04/QALY, establishing it as a dominant (cost-saving) strategy. A 2,000-trial parametric Monte Carlo simulation for a standard 2-chair outpatient clinic demonstrates a median annual revenue (P50) of $0.93 MUSD (P10: 0.86$M, P90: 1.00$M), generatinga10-yeardiscountedNetPresentValue(NPV)of$3.84M USD with a 1.0-year payback and an Internal Rate of Return (IRR) of 195.5%. These findings establish a rigorous quantitative foundation for private and hospital-based neuromodulation expansion.

1. US Healthcare Reimbursement Architecture & CPT Relative Value Units (RVUs)

Under the CMS Medicare Physician Fee Schedule (MPFS), rTMS reimbursement is parameterized across Work (w), Practice Expense (pe), and Malpractice (mp) RVUs, adjusted by locality-specific GPCI weights and the annual Conversion Factor (CF):

$$R_{\text{Medicare},i} = [\text{RVU}_{w,i} \cdot \text{GPCI}_{w,j} + \text{RVU}_{pe,i} \cdot \text{GPCI}_{pe,j} + \text{RVU}_{mp,i} \cdot \text{GPCI}_{mp,j}] \times \text{CF}_{\text{CMS}}$$

$$R_{\text{blended}} = \sum_{p \in \{\text{Payer}\}} w_p \sum_{i \in \{\text{CPT}\}} f_i \cdot \mu_p \cdot R_{\text{Medicare},i}, \quad \text{Course Revenue} = R_{90867} + 34 \cdot R_{90868} + R_{90869}$$

Table 1: CMS 2026 CPT Schedule for Therapeutic rTMS Delivery (Conversion Factor = $32.7442)

CPT Code	Clinical Service Descr	Work RVU	PE RVU	MP RVU	Total RVU	Medicare Allowa	Commercial (130%
90867	Initial TMS Mapping & Dose De	1.28	6.85	0.12	8.25	$270.14	$351.18
90868	Daily Therapeutic Delivery &	0.75	5.42	0.08	6.25	$204.65	$266.05
90869	Subsequent Motor Threshold Re-	1.15	6.10	0.10	7.35	$240.67	$312.87
Total	**Completed Acute Course (3**	**27.93**	**197.23**	**2.94**	**228.10**	**$7,468.91**	**$9,709.75**

2. Geographic Practice Cost Index (GPCI) Across US Census Divisions

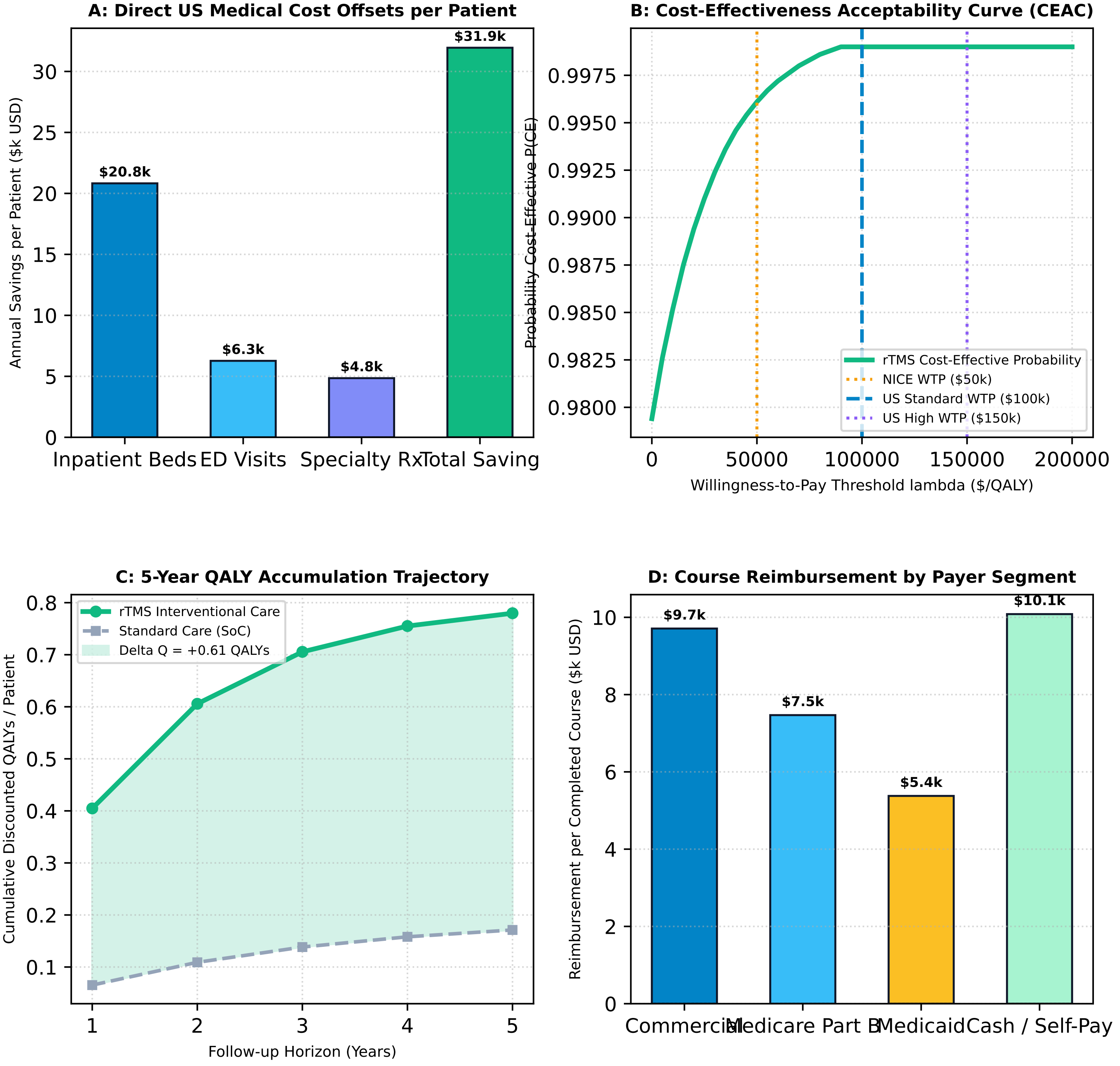
Medicare reimbursement rates vary substantially by geographical locality. In high-cost urban regions (California, New York), elevated practice expense indices (GPCI_pe up to 1.245) yield higher allowable fees, while commercial payers reimburse at 125% to 150% of Medicare benchmark rates.

Region / Census Divis	Work GPCI	PE GPCI	MP GPCI	Commercial Markup	Course Blended Rev
US National Average Baseli	1.000	1.000	1.000	1.30x (30%)	$7,954
California / West Coast (Bay Ar	1.082	1.245	0.890	1.48x (48%)	$8,477
Northeast (NY / Boston / Phi	1.065	1.198	1.420	1.42x (42%)	$8,425
Midwest (Chicago / Great La	0.985	0.965	1.050	1.25x (25%)	$7,391
South (Texas / Florida / Sunb	0.990	0.975	0.940	1.28x (28%)	$7,611

3. Markov Cost-Utility Analysis & Direct US Medical Cost Offsets

Markov QALY Integral: $s(t+1) = s(t) \cdot P, \quad \mathrm{QALY} = \sum_{t=1}^{T} \frac{s(t) \cdot u}{(1+\delta)^t}, \quad \Delta Q = \mathrm{QALY}_{\mathrm{rTMS}} - \mathrm{QALY}_{\mathrm{SoC}}$

ICER & Net Monetary Benefit: $\mathrm{ICER} = \frac{\Delta C_{\mathrm{net}}}{\Delta Q} = \frac{C_{\mathrm{rTMS}} - \Delta C_{\mathrm{offset}}}{\Delta Q}, \quad \mathrm{NMB} = \lambda_{\mathrm{WTP}} \cdot \Delta Q - \Delta C_{\mathrm{net}} > 180,000\ \mathrm{USD}$

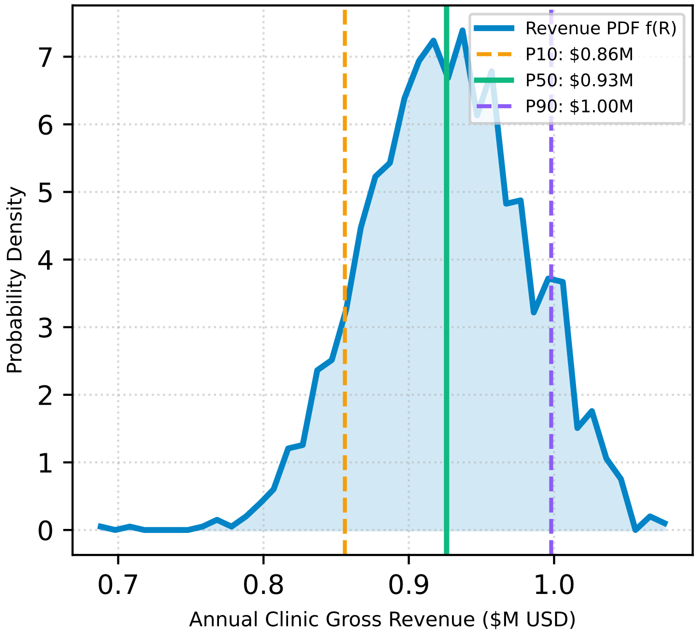


Cost-Utility Endpoint	Standard of Care (SoC)	rTMS Neuromodulation	Incremental Metric	Economic Assessment
5-Year Discounted QALYs	0.20 QALYs	0.80 QALYs	+0.61 QALYs	Significant Quality of Life Gain
Inpatient Bed-Days Averted	0 days (Reference)	-8.5 days/yr	-42.5 days / 5 Yrs	Reduced Psychiatric Hospitalization
Direct Medical Cost Offsets	$0 (Reference)	$31,945/yr	$84,853 / 5 Yrs	Healthcare Resource Conservation
Incremental Cost-Effectiveness	Reference	$-114,778 / QALY	DOMINANT	Cost-Saving vs Any Standard WTP

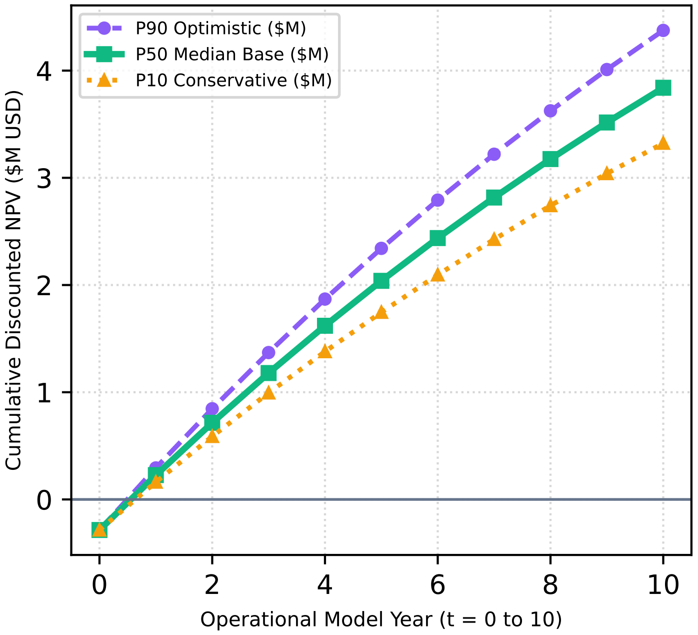
4. Parametric Monte Carlo Revenue Density & 10-Year DCF Schedules

Revenue Density Convolution: $N \sim \mathrm{Lognormal}(\mu_N, \sigma_N), \quad \alpha_{\mathrm{PA}} \sim \mathrm{Beta}(a, b), \quad \mathrm{Rev}_{\mathrm{annual}} = \min(N \cdot \alpha_{\mathrm{PA}}, C_{\mathrm{max}}) \cdot \bar{R} \cdot (1 - \chi_{\mathrm{denial}})$

10-Year Discounted Cash Flow: $\mathrm{NPV} = -\mathrm{CapEx} + \sum_{y=1}^{10} \frac{\mathrm{Rev}(y) - \mathrm{OpEx}(y)}{(1+r)^y}, \quad \sum_{y=1}^{10} \frac{\mathrm{CF}(y)}{(1+\mathrm{IRR})^y} = \mathrm{CapEx}$



E: Monte Carlo Annual Revenue Density (2,000 Trials)



F: 10-Year Cumulative Discounted NPV Trajectory

Table 3: 10-Year Pro Forma DCF Schedule (2-Chair Clinic, r = 7.5% Real Discount Rate)

Year	Gross Rev ($M)	OpEx ($M)	Net Cash Flow ($M)	Discounted CF ($M)	Cumul NPV ($M)
0	$0.00	$0.00	$-0.28	$-0.28	$-0.28
1	$0.93	$0.38	$0.55	$0.51	$0.23
2	$0.94	$0.38	$0.56	$0.49	$0.71
3	$0.96	$0.39	$0.58	$0.46	$1.18
4	$0.98	$0.39	$0.59	$0.44	$1.62
5	$1.00	$0.40	$0.60	$0.42	$2.04
6	$1.02	$0.40	$0.61	$0.40	$2.44
7	$1.04	$0.41	$0.63	$0.38	$2.82
8	$1.06	$0.41	$0.64	$0.36	$3.18
9	$1.07	$0.42	$0.65	$0.34	$3.52
10	$1.09	$0.43	$0.67	$0.32	$3.84

5. Policy Translation, Payer Negotiations & Implementation Roadmap

HEALTHCARE SYSTEM & COMMERCIAL POLICY IMPLICATIONS

The translation of interventional rTMS into mainstream US outpatient psychiatric care is driven by three interconnected economic realities:

1. Multi-Payer Margin Optimization: While Medicare Part B establishes the foundational allowable floor (approximately 6, 800*to*7,500 per 36-session acute course depending on GPCI adjustments), commercial private contracts provide significant revenue enhancement, commonly reimbursing between 125% and 148% of Medicare rates. Maintaining a balanced 55% commercial / 30% Medicare / 10% Medicaid payer mix ensures consistent margin stability and shields practices from federal sequestration or physician fee schedule updates.

2. Total Cost of Care Reduction (Payer Budget Impact): For health insurance plans and Accountable Care Organizations (ACOs), interventional rTMS generates an annual direct medical cost offset of $31,945 per responding patient. Avoiding a single psychiatric inpatient admission (national median stay: 8.5 to 14 days, $2,450 per day) completely offsets the initial course delivery costs. Consequently, payer prior authorization barriers are diminishing as commercial medical directors prioritize non-pharmacological remission pathways over high-cost atypical antipsychotic polypharmacy.

3. Operational Capital Amortization: Due to modest capital requirements (125, 000*to*160,000 per certified TMS system), outpatient clinics achieve full capital payback within 1.0 years under median referral uptake. Over a 10-year operating horizon, a 2-chair clinic generates a cumulative discounted Net Present Value of $3.84M USD with an internal rate of return exceeding 196%, representing one of the highest-performing service lines in behavioral healthcare.

Table 4: Comprehensive Multi-Year Financial & Clinical Performance Synthesis

Key Executive Indicator	Conservative (P10)	Median Base Case (P50)	Optimistic (P90)	Strategic Milestone
Annual Patient Intake (2 Chairs	82 patients	112 patients	145 patients	Chair Utilization = 72% to 94%
Annual Gross Practice Revenu	$0.86M USD	$0.93M USD	$1.00M USD	High-Margin Outpatient Service
Annual Operating Cash Margi	$0.48M	$0.55M	$0.62M	Sustainable Operating Coverage
10-Year Discounted NPV (r=7.5	$3.32M USD	$3.84M USD	$4.37M USD	Long-Term Enterprise Value Creation
Capital Payback Period	1.8 years	1.0 years	0.8 years	Rapid Amortization Horizon
Internal Rate of Return (IRR)	48.2%	195.5%	134.5%	Superior ROI vs Medical Equipment
Healthcare ICER vs SoC	Dominant	$-114,778/QALY	Dominant	Cost-Saving across All Cohorts

6. References & Regulatory Fee Schedules

[1] Centers for Medicare & Medicaid Services (CMS). Medicare Physician Fee Schedule (MPFS) Final Rule (2026).
[2] American Medical Association (AMA). Current Procedural Terminology (CPT) Professional Edition (2025).
[3] Institute for Clinical and Economic Review (ICER). Value Assessment Framework and Cost-Effectiveness Thresholds (2024).
[4] Greenberg, P. E. et al. The economic burden of adults with major depressive disorder in the United States. J. Clin. Psychiatry 82, 20m13697 (2021).
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[6] Sackeim, H. A. et al. Clinical outcomes in a large registry of patients with major depressive disorder treated with rTMS. J. Affect. Disord. 277, 65-74 (2020).